

Total No. of Questions : 8]

SEAT No. :

PC2491

[Total No. of Pages : 2

[6354]-620

B.E. (Mechanical Engineering)

PRODUCT DESIGN AND DEVELOPMENT

(2019 Pattern) (Semester - VII) (Elective - IV) (402045 A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

- Q1)** a) Explain Pugh's chart with example. [7]
- b) Write note on Economic Analysis. [6]
- c) Write a short note on SWOT analysis for a selection of profitable product. [4]

OR

- Q2)** a) Explain Product Characteristics. [7]
- b) Describe in detail Concept Selection Process. [6]
- c) Write note on Alternate thinking in idea generation approach. [4]

- Q3)** a) Describe in detail Design review/part Print Analysis. [7]
- b) Explain Qualitative & Quantitative Economic Analysis. [6]
- c) Write a short note on Arrangement of Dimensions. [4]

OR

P.T.O.

- Q4)** a) Explain BOM with example. [7]
b) Explain Value analysis in detail. [6]
c) What is Fit? Describe the types of Fits. [4]

- Q5)** a) Explain production Part Approval Process tools in detail. [8]
b) Define Purchase Order and Product Costing in vendor development. [6]
c) Explain Drawing office procedure in detail. [4]

OR

- Q6)** a) Describe in detail Legal Product of design patents. [8]
b) Explain Product Testing and its Validation. [6]
c) Describe Organization Structure in detail. [4]

- Q7)** a) Write a short note on Advance Product Quality Planning. [8]
b) What is the importance of PLM and PDM for a Designer? [6]
c) Enlist Industrial Safety to be consider in Product Design. [4]

OR

- Q8)** a) What is PLM? What are the phases involved in it? [8]
b) Explain Aesthetic and aesthetics consideration in product design. [6]
c) What are guidelines for Design for Robustness? Discuss. [4]

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